

Final Project Summary

Hotel Booking Pricing Analysis: Customer Retention and Dynamic Pricing

Project Overview

The Hotel Booking Pricing Analysis project was conducted to examine booking patterns, pricing behavior, customer retention, and cancellation trends within the hospitality industry. Using the hotel booking dataset, the project applied data cleaning, exploratory data analysis (EDA), statistical testing, customer segmentation, and business intelligence techniques to generate actionable insights for revenue management.

The primary objective was to identify the key factors influencing Average Daily Rate (ADR), customer cancellations, and booking behavior, enabling data-driven decisions that improve revenue performance and customer retention.

Methodology

The project was completed through the following phases:

1. Data Cleaning and Preparation

- Handled missing values in critical columns such as agent, company, children, and country.
- Removed duplicate records.
- Treated ADR outliers using the Interquartile Range (IQR) method.
- Validated data quality and prepared the dataset for analysis.

2. Exploratory Data Analysis (EDA)

- Analyzed booking trends and seasonal demand patterns.
- Examined ADR distribution across hotel types and customer segments.
- Investigated customer demographics and booking behavior.
- Identified relationships between pricing variables and booking outcomes.

3. Statistical Analysis

- Performed Chi-Square tests to evaluate relationships between categorical variables and booking cancellations.
- Conducted T-Tests to compare ADR between canceled and non-canceled bookings.
- Validated statistically significant factors affecting customer behavior and pricing.

4. Customer Segmentation

- Segmented customers based on booking lead time, stay duration, booking characteristics, and repeat guest status.
- Identified high-value and high-risk customer groups.

5. Dashboard Development

- Built an interactive Power BI dashboard to visualize key revenue management metrics.
 - Implemented filters, slicers, and drill-down capabilities for business users.
 - Consolidated project findings into a business-facing reporting solution.
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Key Findings

Pricing Drivers

The analysis identified several variables that significantly influence ADR, including:

- Lead Time
- Hotel Type
- Arrival Month
- Market Segment
- Distribution Channel
- Room Type
- Customer Type
- Booking Changes

Customer Cancellation Patterns

- Bookings with longer lead times exhibited higher cancellation rates.
- Cancellation behavior varied significantly across market segments and customer types.
- Certain distribution channels experienced more frequent cancellations.

Customer Retention Insights

- Repeat guests demonstrated lower cancellation rates and greater booking consistency.
- High-value customer segments generated stronger revenue performance.

Seasonal Demand Trends

- Demand and ADR fluctuated across different months and seasons.
 - Peak demand periods presented opportunities for revenue optimization through dynamic pricing.
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Business Recommendations

1. Implement dynamic pricing strategies based on demand patterns, seasonality, and booking lead time.
 2. Develop targeted retention programs for repeat and high-value customers.
 3. Reduce cancellation rates through optimized booking policies and customer engagement initiatives.
 4. Continuously monitor revenue and booking KPIs through interactive dashboards.
 5. Optimize distribution channel performance to maximize revenue and reduce booking risks.
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Deliverables Completed

- Cleaned and validated hotel booking dataset.
 - Exploratory Data Analysis (EDA) notebooks and visualizations.
 - Statistical testing and hypothesis validation.
 - Customer segmentation analysis.
 - Interactive Power BI dashboard.
 - GitHub repository with documentation and reproducible notebooks.
 - Business presentation and strategic recommendation report.
 - Comprehensive README documentation.
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Business Impact

This project provides a data-driven framework for improving hotel revenue management by identifying critical pricing drivers, customer retention opportunities, and cancellation risk factors. The insights generated can support strategic pricing decisions, enhance customer loyalty initiatives, and improve overall business performance through more effective revenue optimization practices.

Conclusion

The Hotel Booking Pricing Analysis project successfully transformed raw booking data into actionable business intelligence. Through analytical techniques and dashboard-driven reporting, the project identified key revenue drivers, customer behavior patterns, and operational opportunities. The resulting recommendations provide a foundation for informed decision-making and sustainable revenue growth within the hospitality industry.